

Android Assignment

Timebox

6–7 hours (please do not exceed this)

Tech Stack

- Language: **Kotlin**
- UI: **Jetpack Compose**
- Architecture: **ViewModel + UI State**
- Coroutines
- Any IDE (Android Studio preferred)

Tooling Policy

You **may use GPT / Docs / StackOverflow**.

However, the goal is to demonstrate **understanding**, not copy-paste.

Problem Statement

Build a **passwordless authentication flow** using **Email + OTP**, followed by a **session screen** that tracks login duration.

No backend is required - all logic can be implemented locally.

Functional Requirements

1 Email + OTP Login

- User enters an **email address**
- Taps **Send OTP**
- A **6 digit OTP** is generated locally
- User enters OTP to log in

2 OTP Rules (Important)

- OTP length: **6 digits**
- OTP expiry: **60 seconds**

- Maximum attempts: **3**
- Generating a new OTP:
 - Invalidates the old OTP
 - Resets attempt count

You must store OTP data **per email**.

3 Session Screen

After successful login:

- Show **session start time**
- Show **live session duration** (mm:ss)
- Provide a **Logout** button

The timer:

- Must survive recompositions
- Must stop correctly on logout

4 External SDK Integration (Mandatory)

Integrate **one** of the following:

- Firebase Analytics
- Sentry
- Timber

You must:

- Add dependency
- Initialize correctly
- Log the following events:
 - OTP generated
 - OTP validation success
 - OTP validation failure
 - Logout

Technical Expectations

You are expected to demonstrate understanding of:

Jetpack Compose

- `@Composable`
- `remember`
- `rememberSaveable`
- `LaunchedEffect`
- State hoisting
- Recomposition handling

Architecture

- ViewModel usage
- One-way data flow
- Clear separation of UI & business logic

Kotlin / Data Structures

- Appropriate use of collections (e.g., `Map`)
- Time-based logic
- Defensive coding

Suggested Project Structure (not mandatory)

ui/

```
|— LoginScreen.kt
|— OtpScreen.kt
|— SessionScreen.kt
```

viewmodel/

```
|— AuthViewModel.kt
|— AuthState.kt
```

data/

```
|— OtpManager.kt
```

analytics/

```
|— AnalyticsLogger.kt
```

Edge Cases to Handle

- Expired OTP
- Incorrect OTP
- Exceeded OTP attempts
- Resend OTP flow
- Screen rotation should not break state

Documentation (Mandatory)

Include a **README.md** explaining:

1. OTP logic and expiry handling
2. Data structures used and why
3. Which external SDK you chose and why
4. What parts you used GPT for vs what you understood and implemented yourself

What Not to Do

- No global mutable state
- No UI logic inside ViewModel
- No blocking calls on main thread
- No copying full projects/templates

Bonus (Optional)

- Visual countdown timer
- Sealed UI states
- Retry cooldown handling

Submission Instructions

- Share a **GitHub repository link**
- Include setup instructions
- App should build and run without extra steps

Evaluation Criteria

We will evaluate:

- Code clarity & structure

- State management correctness
- Compose usage
- Data structure reasoning
- SDK integration correctness
- README quality

Note

This assignment intentionally allows AI assistance.

We are evaluating **how well you understand and apply concepts**, not how fast you generate code.